The Cost of Inaction

Adolescent Girls' Education and Human Capital in Nepal

A policy analysis using the Nepal Living Standards Survey IV (2022/23)

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Executive Summary

Nepal has made a great progress in getting girls into school. Today, adolescent girls are enrolled at higher rates than boys (40.2 per cent versus 34.8 per cent). Yet this progress in the classroom does not carry through to the labour market. Nearly half of adolescent girls aged 15–24 (46.8 per cent), are not in education, employment, or training (NEET), almost double the rate for boys (26.4 per cent). The gender gap in Nepal is no longer primarily a problem of school access, it is a problem of the transition from school to economic opportunity.

This analysis quantifies the economic cost of that lost transition. Using nationally representative microdata from the Nepal Living Standards Survey IV (NLSS IV, 2022/23) and a combination of econometrics and causal machine-learning, we estimate that Nepal forgoes approximately NPR 8.1 billion (about USD 61 million) in lifetime productivity for every single cohort of adolescent girls who do not complete secondary education. A new cohort reaches this stage every year, so this is a recurring annual loss, not a one-off. To place that figure in perspective, this is equivalent to roughly 9.4 per cent of Nepal's annual health-sector budget, 7.6 per cent of the social security allowance budget, and 4.0 per cent of the education-sector budget.

Three policy insights stand out from these findings:

1. **Child marriage is the single most powerful driver of exclusion.** Being currently married is associated with a 61-percentage-point fall in enrolment and a 42-point rise in NEET, larger than the combined association of poverty, caste, and disability. The share of girls who are married jumps from 10% at ages 15–18 to 56% at ages 19–24.
2. **Secondary education has a robust, stable effect on inclusion.** For girls and women, completing secondary school is associated with a 13.2-percentage-point reduction in the probability of being NEET, and this effect is stable across all seven provinces and all caste groups. The parallel estimate for boys and men is smaller (approximately 7 percentage points), as male exclusion is shaped more by labour-demand conditions than by individual schooling. The parallel estimate for boys and men is smaller (approximately 7 percentage points), as male exclusion is shaped more by labour-demand conditions than by individual schooling.
3. **The returns to schooling rise steeply with each level.** Each additional year of schooling is associated with a 14.5 per cent average increase in earnings for girls, and these returns are convex, rising from about 8% at primary, to 27% at secondary, and 35% at higher secondary.

The policy implication is clear: Nepal's next frontier is not only to get girls into school, but to keep them in school long enough for education to change their life trajectories. Investments need to reach girls before the post-18 transition, when the risks of marriage, dropout, and labour-market exclusion accelerate sharply. This means rewarding every additional year of schooling retained, with particular emphasis on lower and upper secondary, where the risks of dropout concentrate and the returns are largest and higher secondary completion.

At the same time, education policy cannot be separated from child marriage prevention. Reducing child marriage is not only a protection objective, it is also central to education, health — early pregnancy and maternal mortality are closely linked to child marriage —, employment, and economic growth. The findings show that marriage is one of the strongest predictors of girls' exclusion from both school and work. Interventions that keep girls in school and prepare them with skills for work and life, delay marriage,

support families, and expand adolescent girls' aspirations should therefore be treated as mutually reinforcing components of the same investment agenda.

Finally, secondary education must be paired with credible pathways into economic opportunity. Girls need not only to complete school, but also to see a viable transition into skilling, employment, entrepreneurship, and further learning. Without this, Nepal risks sustaining high levels of female school enrolment while continuing to lose the economic returns of girls' education.

1. Introduction and Motivation

Investing in girls' education is among the most widely-evidenced development priorities in the world. The case is usually made in terms of rights and long-run social benefits. This report makes the case in the language of public finance: what does Nepal lose, in measurable economic terms, when adolescent girls do not complete secondary education and do not transition into productive economic activity?

The motivation is both moral and fiscal. Nepal's Constitution guarantees education as a fundamental right, and the country has committed to the Sustainable Development Goals. But for a Ministry of Finance weighing competing claims on a constrained budget, the relevant question is concrete: is closing the gender gap in the school-to-work transition an expenditure, or an investment with a positive return? This analysis is designed to answer that question with defensible numbers.

This work follows the methodology of UNICEF's cost-of-inaction analyses in other contexts —notably the *Cost of Inaction: Girls' Education in Afghanistan* (2022) and Nepal's own *Cost of Inaction of Child Marriage* (2014)— while adapting the approach to Nepal's current data and using more recent causal-inference techniques.

2. Data

The analysis uses the Nepal Living Standards Survey IV (NLSS IV), 2022/23, produced by the National Statistics Office (NSO) of Nepal. The NLSS IV is a nationally representative, two-stage stratified household survey covering all seven provinces. It interviewed 46,870 individuals in 9,600 households.

The core analytical sample is 9,785 youth aged 15–24 (4,919 girls and 4,866 boys). For the econometric analyses we use the broader sample of 13,586 women aged 15–64. All estimates use the survey's sampling weights and account for its complex design (primary sampling units and provincial strata).

Key variables include school enrolment and attainment (NLSS education module), employment and NEET status (defined using the ILO seven-day reference period), monthly wages, marital status, caste and ethnicity (grouped as Brahmin/Chhetri, Janajati, Dalit, and Religious/Linguistic minorities), disability status (Washington Group questions), wealth quintile, and province of residence.

All code, analytical scripts, and documentation underlying this report are publicly available at https://github.com/hgrueso/nepal-cost-of-inaction. The repository contains the full data-cleaning, regression, causal-forest, and cost-of-inaction pipeline in R, together with the source documentation for every external figure used (including the government budget comparisons). The underlying NLSS IV microdata are not redistributed and must be obtained directly from the National Statistics Office of Nepal, in line with its data-access terms.

3. Methodology and Key Assumptions

The analysis combines three methodological approaches designed to measure gaps, estimate the returns to education, and value lost earnings. Below is a summary of the key aspects of these methods, full technical details are included in the Annex.

3.1 Measuring the gaps

We first document the size and structure of the gender gap using descriptive statistics and a multivariate (linear probability) regression that isolates the independent association of each factor (marriage, caste, wealth, disability, geography) with enrolment and NEET status.

3.2 Estimating the effects of education

To estimate the effect of secondary education on the probability of being NEET, we use a causal forest (a machine-learning method for estimating causal effects, Wager and Athey 2018). This approach allows the effect of education to vary across individuals and groups, and provides a credible average treatment effect that is more robust than a simple correlation.

3.3 Valuing the lost earnings

To translate education into economic value, we estimate a Mincer wage equation, the standard tool in labour economics relating wages to years of schooling and experience. Because wages are observed only for women in paid employment (about 25% of working-age women), we test for selection bias using a Heckman selection model. The selection term is statistically significant but changes the estimated return by less than two percentage points, confirming that the wage estimates are robust (see Annex).

3.4 Key assumptions and limitations

Three assumptions are central to interpreting the cost-of-inaction figure. For each, we state the assumption plainly, explain the direction of any bias it could introduce, and describe what we did to test or correct for it.

- **Formal-sector focus.** Wage returns are estimated on women in paid employment, who are a minority (about 25%) of all working-age women. Therefore, the estimated return could be biased if the women who work for wages differ systematically from those who do not. We test for this directly using a Heckman (1979) two-step selection model, which models the probability of entering wage employment (using household composition, the presence of young children, and wealth as factors affecting participation but not the wage itself) and corrects the wage equation accordingly. The selection term is statistically significant (correlation $\rho = 0.34$, $p < 0.01$), confirming that positive selection exists. However, correcting for it shifts the estimated return by less than two percentage points. The wage estimates are therefore robust to selection bias, and the analysis captures the economic-productivity loss associated with formal labour-market engagement. Full detail is in Annex A.2.
- **Labour-market entry.** The break-even and cost figures assume that a girl who completes secondary education enters the labour market and earns the median wage. In reality, not all will, so the figures describe potential economic value rather than a guaranteed return. We mitigate the resulting overstatement in two ways: by using a deliberately conservative wage-return benchmark (see below), and by stating the per-girl break-even as a theoretical maximum spend rather than a

prediction. The figure should be read as the economic value at stake if the transition to work were achieved, not as a forecast of realised earnings.

- **An upper bound, not a lower bound.** The estimate assumes that a girl who leaves NEET status begins earning the median wage immediately. In practice, this is unlikely to be fully achievable because girls and young women may face barriers such as job search frictions, childcare responsibilities, mobility constraints, and restrictive social norms. For this reason, the estimated recoverable earnings should be interpreted as an upper bound on what could realistically be recovered, not as a conservative minimum. The conservative element of the analysis comes from the wage-return benchmark used to estimate earnings, rather than from the assumption that girls can transition immediately into paid work.

To further guard against overstatement, the headline cost calculation uses a deliberately conservative wage-return benchmark of 9% per year of schooling, based on the International Labour Organization's standard benchmark, rather than the returns estimated directly in this analysis. As shown in the Annex, this 9% benchmark is lower than even the estimated return to primary schooling, and substantially below the estimated returns to secondary and higher secondary education, which are most relevant to the policy case. The resulting cost figure should therefore be interpreted as conservative with respect to expected earnings, even though the transition from NEET into paid work is modelled as an upper-bound scenario.

4. Findings

4.1 The gap is in the labour market, not the classroom

Girls in Nepal now enrol in school at higher rates than boys, but are far more likely to be excluded from both school and work as they reach adulthood. The female NEET rate of 46.8% is nearly double the male rate, and the gap widens with age, from about 9 percentage points at ages 15–18 to almost 25 points at ages 19–24.

Figure 1. *Enrolment, employment, and NEET by sex, youth aged 15–24*

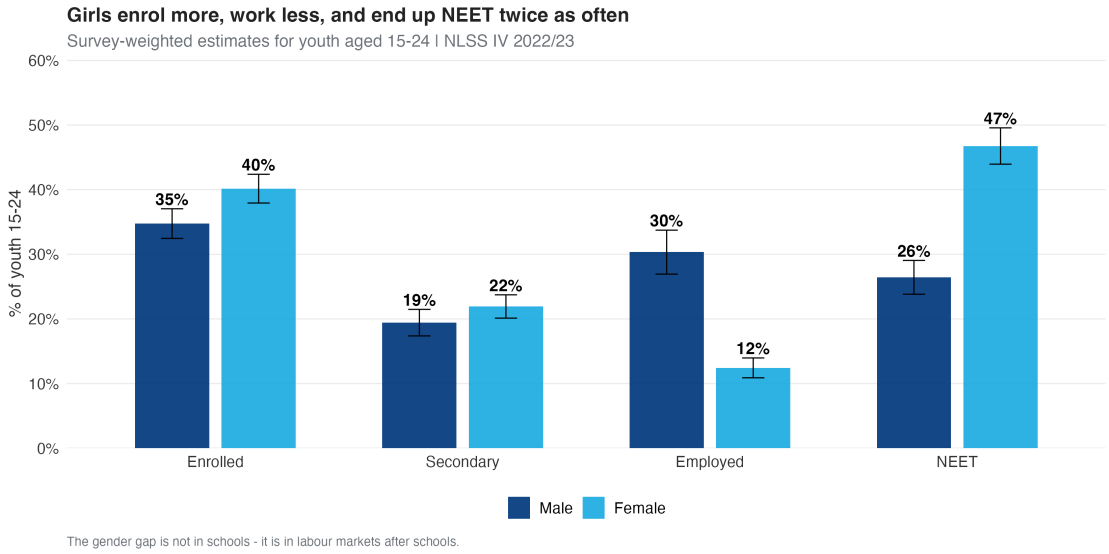
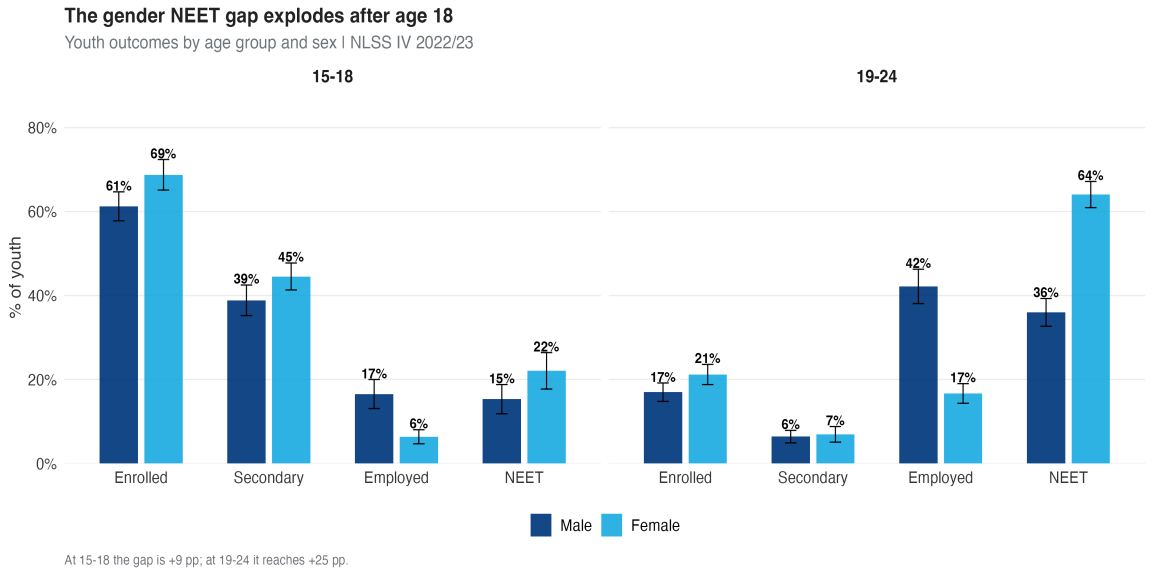


Figure 2. *Youth outcomes by age group and sex. The NEET gap widens sharply after age 18.*



4.2 Child marriage is the strongest driver of exclusion

Among all the factors we examine, marital status has by far the largest association with girls' exclusion. Being currently married is associated with a 61-percentage-point reduction in enrolment and a 42-point increase in NEET, after controlling for poverty, caste, geography, and disability. This pattern holds across every caste group.

Crucially, the prevalence of marriage changes dramatically across the adolescent years: 10 per cent of girls aged 15–18 are married, rising to 56% by ages 19–24. The divergence in outcomes happens at this transition, which is why interventions need to reach girls before they reach it.

Figure 3. *NEET rate by marital status and caste, girls 15–24. Marriage is associated with near-universal NEET across all groups.*

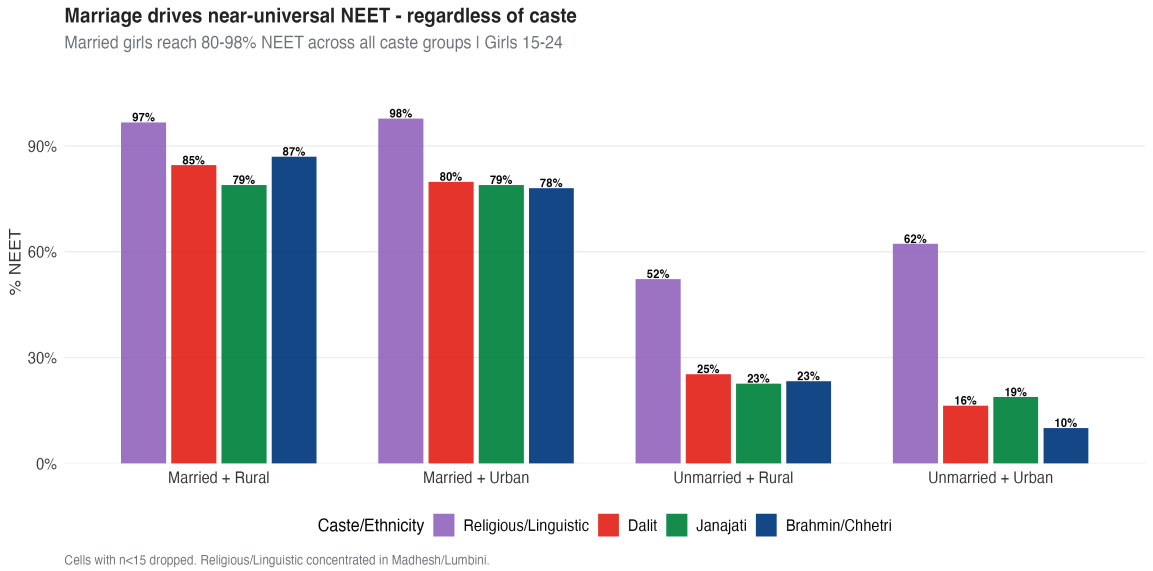
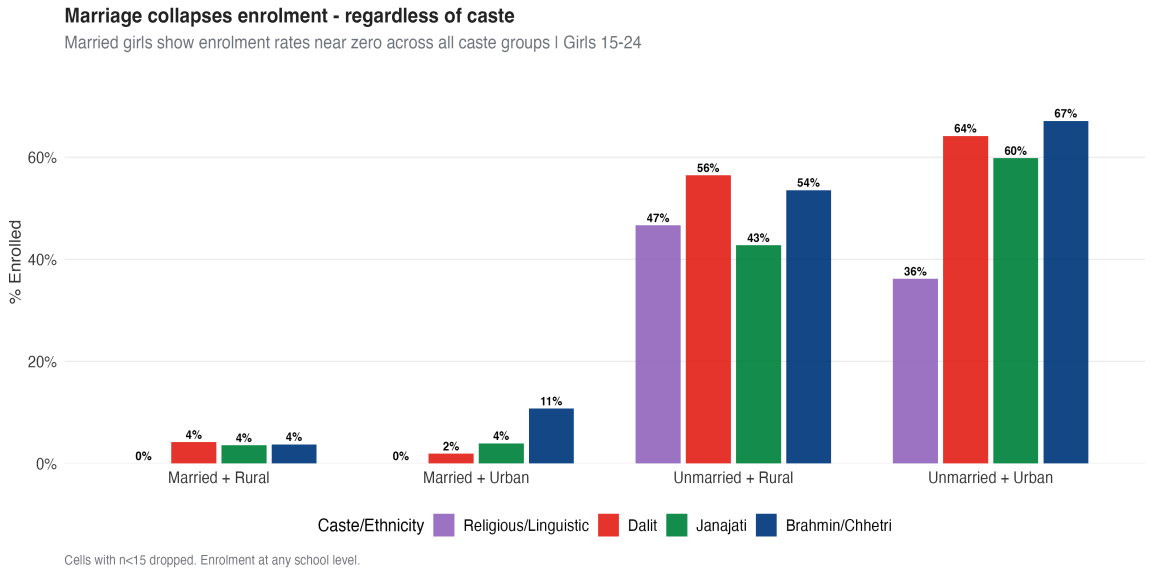


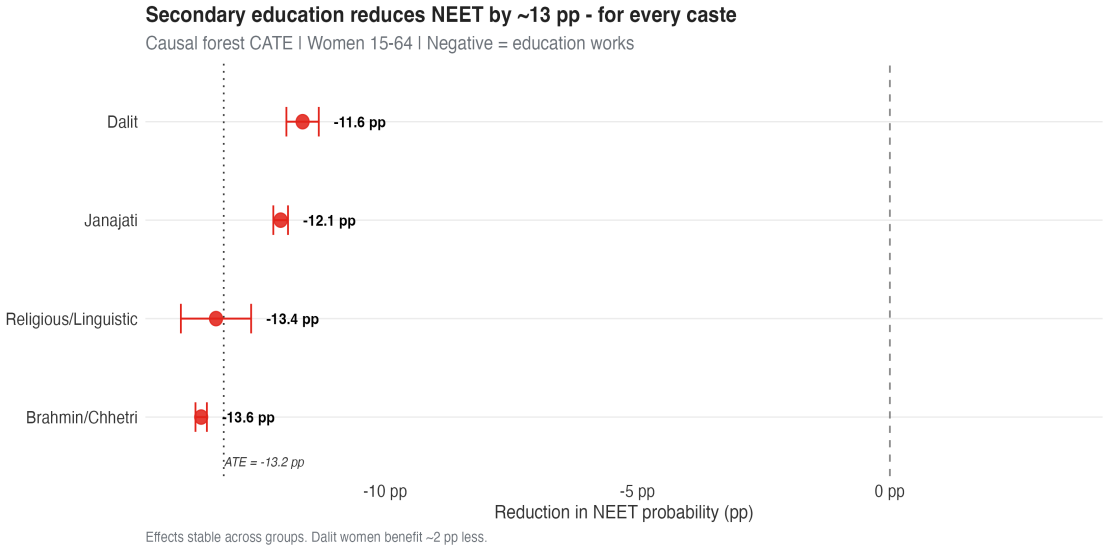
Figure 4. *Enrolment by marital status and caste, girls 15–24*



4.3 Education works everywhere

The causal-forest analysis shows that completing secondary education is associated with a 13.2-percentage-point reduction in the probability of being NEET for women aged 15–64. This effect is remarkably stable: it holds across all seven provinces and all caste groups, indicating that education is a reliable lever for inclusion regardless of where a girl lives or which community she belongs to.

Figure 5. *Estimated effect of secondary completion on NEET probability, by caste*

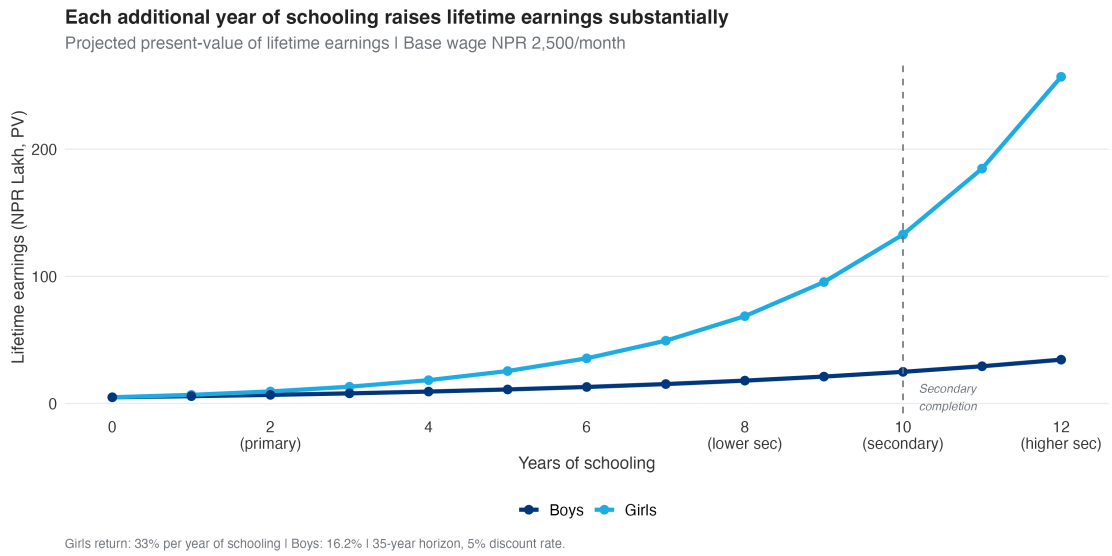


4.4 The returns to schooling rise with each level

Each additional year of schooling is associated with a 14.5 per cent average increase in girls' lifetime earnings. Critically, these returns are convex, meaning that they rise with the level of education, from roughly 8% at primary, to 27% at secondary, to 35% at higher secondary. This means the economic loss

from a girl leaving school is greatest precisely at the upper-secondary level, where Nepal's gender gap in the school-to-work transition is widening.

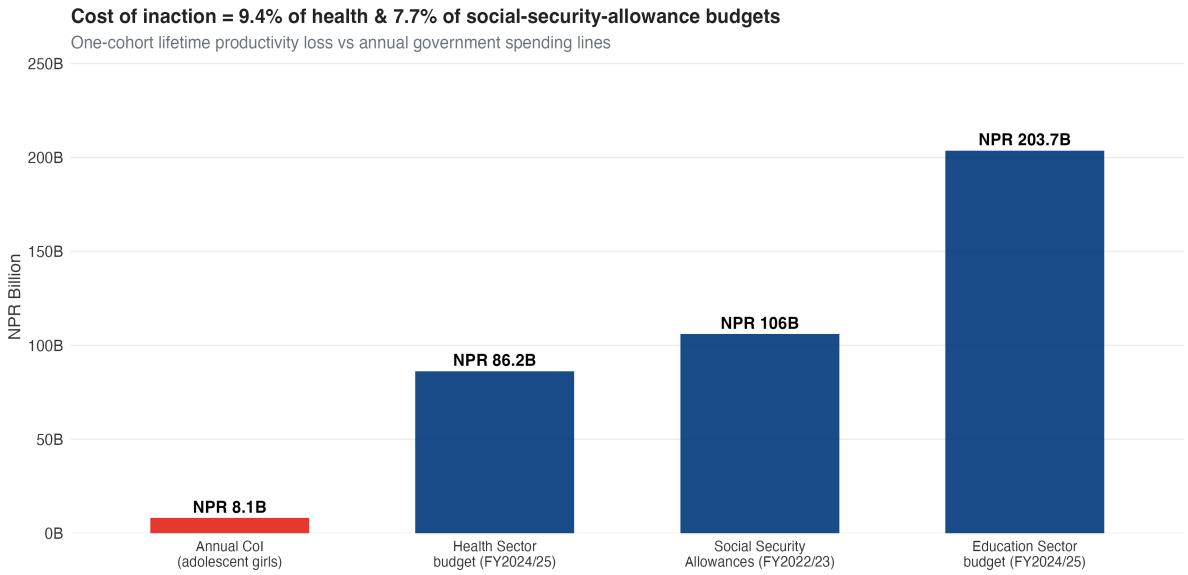
Figure 6. *Projected lifetime earnings by years of schooling and sex.*



4.5 The cost of inaction

Combining the causal effect of education, the conservative wage return, and the size of the affected population (680,231 girls), we estimate that Nepal forgoes approximately **NPR 8.1 billion (about USD 61 million)** in lifetime productivity for each cohort of adolescent girls who do not complete secondary education.

Figure 7. *The annual cost of inaction compared with national budget lines*



Sources: This analysis (NLSS IV 2022/23); health budget MoHP FY2024/25; social security allowances UNICEF Nepal FY2022/23; education budget MoF FY2024/25.

This figure is equivalent to about 9.4% of the annual health-sector budget (fiscal year 2024/25), 7.6% of the social security allowance budget, and 4% of the education budget. This is likely an understatement considering that the analysis is not taking into account non-market benefits of girls' education such as lower fertility, better child health, reduced adolescent pregnancy, and intergenerational gains.

5. Policy Recommendations

The findings point to four complementary priorities for investment. Education is necessary, but not sufficient: keeping girls in school will have limited impact unless it is paired with measures that delay marriage and create credible pathways into economic opportunity.

1. **Education — universal secondary completion with year-by-year retention.** Because returns are convex, programmes should reward every year of schooling retained, with the largest gains concentrated at upper-secondary and tertiary level. Conditional support should target retention through the post-18 transition.
2. **Child marriage — treat prevention as an economic priority.** Marriage is the single largest driver of girls' exclusion, and its prevalence rises sharply after age 18. A comprehensive response combines enforcement of existing legislation, community-level social-norm change at scale, and social protection support to relieve the economic pressures that lead families toward early marriage. These efforts, concentrated in the provinces with the widest gaps (Madhesh and Lumbini), should reach girls before the transition.
3. **Livelihoods — bridge programmes, vocational training, and economic opportunity.** Because returns are concentrated in formal wage work, which absorbs only a minority of women, education must be paired with demand-side measures: skills bridges from school to work, vocational training, and second-chance pathways for those who have already left.
4. **Equity — close the gaps facing Dalit girls.** Even after completing the same level of education as other groups, Dalit women still face a labour-market disadvantage of about 2 percentage points, a gap that education alone does not close. Pair education investment with additional support such as apprenticeships, targeted scholarships, and anti-discrimination measures.

Taken together, these investments are strongly cost-effective. Any secondary-completi beon programme costing less than the per-girl break-even (on the order of USD 90 in cumulative lifetime social spending) would yield a positive economic return in pure productivity terms, before counting the substantial health, fertility, and intergenerational benefits that this analysis does not monetise.

Annex: Technical Detail

This annex documents the three methods used in the analysis, the formal model behind each, and the rationale for the key parameter choices. The methods are complementary: the causal forest (A.1) identifies *whether and for whom* education reduces exclusion; the Mincer / Heckman framework (A.2) values the *earnings* associated with education and tests it for selection bias; and the cost-of-inaction calculation (A.3) combines the two with population and discounting assumptions to produce the headline figure.

A.1 Causal Forest: The effect of education on exclusion

We estimate the effect of secondary completion on the probability of being NEET (not in education, employment, or training) using a generalised random forest / causal forest (Wager and Athey, 2018). Writing $Y_i(1)$ and $Y_i(0)$ for the potential NEET outcomes of woman i with and without secondary completion, the target parameter is the average treatment effect:

$$\tau_{ATE} = \mathbb{E}[Y_i(1) - Y_i(0)]$$

Because each woman is observed in only one state, the forest estimates the conditional average treatment effect (CATE) as a function of covariates x:

$$\tau(x) = \mathbb{E}[Y_i(1) - Y_i(0) \mid X_i = x]$$

It does so by growing an ensemble of "honest" regression trees that partition the covariate space to maximise heterogeneity in the treatment effect, then averaging across trees. Identification rests on the unconfoundedness (selection-on-observables) assumption:

$$(Y_i(1),\ Y_i(0)) \perp W_i \mid X_i,$$

where $W_i \in \{0,1\}$ indicates secondary completion. In our implementation:

- **Sample:** women aged 15–64 with non-missing treatment and outcome (n = 13,586).
- **Treatment W:** completed secondary education.
- **Outcome Y:** NEET status (also estimated for employment and wage employment).
- **Covariates X:** age, age², rural residence, poverty, wealth quintile, caste/ethnicity, province, disability, and marital status.
- **Tuning:** 2,000 trees, survey-weighted (sampling weights), all parameters tuned by cross-validation, fixed seed for reproducibility.

Result: the estimated ATE is −13.2 percentage points (secondary completion is associated with a 13.2-point lower probability of being NEET) and is stable across all seven provinces and all caste groups.

Rationale. A causal forest is preferred over a single linear regression for two reasons. First, it relaxes the assumption that the effect of education is the same for everyone: it estimates how the effect varies across provinces and caste groups, which is exactly the heterogeneity policymakers need to target investment. Second, the "honesty" property (using separate data to choose the tree splits and to estimate effects within them) produces valid confidence intervals and reduces overfitting. The design controls for confounding from observed characteristics; because it is not a randomised experiment, it does not eliminate bias from unobserved factors, so we interpret it as the best available observational estimate rather than a definitive causal parameter.

A.2 Mincer wage equation and Heckman selection correction

The wage equation. Returns to schooling are estimated with a Mincerian earnings function (Mincer, 1974), augmented with a quadratic schooling term to capture the convexity of returns observed for women:

$$ln(w_i) = \beta_0 + \beta_1 S_i + \beta_2 S_i^2 + \beta_3 X_i + \beta_4 X_i^2 + \gamma' \mathbf{Z}_i + \varepsilon_i$$

where w_i is the monthly wage, S_i is years of schooling (centred at the sample mean), X_i is potential experience (age − S_i − 6), and Z_i is a vector of controls (rural residence, caste, province fixed effects). Because schooling is centred, the marginal return at schooling level s is:

$$\frac{\partial\, ln(w_i)}{\partial s_i} = \beta_1 + 2\beta_2 (s - \bar{S})$$

The positive estimated β_2 means returns rise with the level of education: approximately 8% at primary, 27% at secondary, and 35% at higher secondary, with an average marginal return of about 14.5% across the observed distribution. Estimation uses survey weights and heteroskedasticity-robust (HC1) standard errors.

The selection problem. Wages are observed only for the ≈25% of working-age women in paid employment. If unobserved factors that raise the probability of working also raise wages, the OLS estimate of β_1 is biased. We test and correct for this with the Heckman (1979) two-step estimator.

Step 1 — selection equation (probit). We model the probability that a wage is observed:

$$Pr(D_i = 1 \mid V_i) \;=\; \Phi(V_i'\delta)$$

where $D_i = 1$ if a wage is observed and Φ is the standard normal CDF. The vector V_i contains the wage-equation covariates plus exclusion restrictions, variables affecting participation but excludable from the wage: number of children under five, number of other adult women in the household, household size, and wealth quintile. From this we form the inverse Mills ratio:

$$\lambda_i = \frac{\varphi(V_i'\delta)}{\Phi(V_i'\delta)}$$

Step 2 — outcome equation with correction. Adding λ_i to the wage equation:

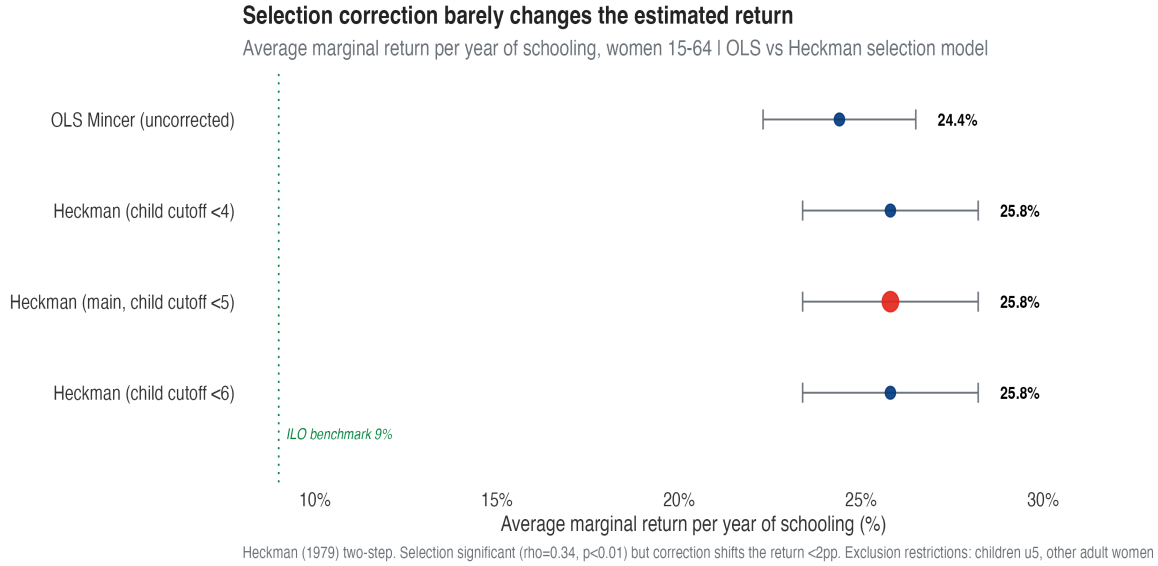
$$ln(w_i) = \beta_0 + \beta_1 S_i + \beta_2 S_i^2 + \beta_3 X_i + \beta_4 X_i^2 + \gamma' \mathbf{Z}_i + \theta \lambda_i + u_i$$

The coefficient $\theta = \rho\sigma$ captures the selection effect; its significance tests for selection bias, and ρ is the correlation between the selection and wage errors.

Result: $\rho = 0.34$ with the inverse Mills ratio significant at $p < 0.01$, there is statistically significant positive selection into wage employment. However, the corrected return differs from the uncorrected estimate by less than two percentage points, and is stable across alternative child-age cutoffs (under 4, 5, 6) in the exclusion restriction. The Mincer estimates are therefore robust to sample selection.

Rationale. The Heckman model is the standard econometric tool for exactly the concern reviewers raised: in a labour market as informal as Nepal's, the women observed earning wages are not a random sample of all women. The exclusion restrictions are chosen from the female-labour-supply literature, the presence of young children and the availability of other adult women in the household shift whether a woman works, but have no direct effect on the wage she would command. This correction addresses selection into *observed wages only*; it is not a causal-identification strategy for schooling itself, which is the role of the causal forest in A.1. The two methods deliberately address different biases.

Figure A1. *Average marginal return to schooling: OLS versus Heckman selection-corrected estimates, with robustness across exclusion-restriction definitions.*



A.3 The cost-of-inaction calculation

The cost of inaction is the present value of lifetime earnings forgone by one cohort of girls who do not complete secondary education. The annual earnings gain from the additional schooling associated with secondary completion is:

$$\Delta w \;=\; w_0 \times (1 + r)^{\Delta S - 1},$$

where w_0 is the baseline median annual wage, r is the per-year return to schooling, and ΔS is the additional years of schooling associated with secondary completion. The lifetime present value over a working horizon of T years, discounted at rate δ, uses the standard annuity factor:

$$PV \;= \Delta w \frac{1 - (1 + \delta)^{-T}}{\delta}$$

The cost of inaction for a population segment combines the share of girls who would no longer be NEET if they completed secondary (the absolute CATE from A.1), the segment population, and the per-girl present value:

$$CoI = |\tau(x)| \times N \times PV,$$

where N is the number of girls in the segment not completing secondary education. National totals are obtained by summing across provinces.

Parameter values (from the analysis code):

- **Return r = 9% per year**. The ILO benchmark, used deliberately instead of our own estimates which yields a more conservative estimate.
- **Years gained ΔS = 2**. The additional schooling associated with secondary completion.
- **Discount rate δ = 5%**. Standard for long-horizon social cost-benefit analysis.

- **Horizon T = 35 years**. An adolescent girl's approximate working life.
- **Population N = 680,231 girls**. Girls aged 15–24 not completing secondary, scaled from NLSS IV survey weights to a national total of ≈1.4 million.
- **Baseline wage w_0**. Province-specific median monthly wage of women in salaried employment.

Result: aggregating across provinces yields a national cost of inaction of approximately NPR 8.1 billion (about USD 61 million) per cohort. This is a recurring annual loss, as a new cohort reaches this stage each year.

Rationale for the conservative choices. Three deliberate choices make the headline figure conservative. (1) The 9% return is below even our estimated primary-school return (≈8%) and far below the secondary and higher-secondary returns (27–35%) that dominate the policy case. (2) The figure counts only formal-sector wage productivity and excludes non-market benefits such as lower fertility, improved child health, reduced adolescent pregnancy, and intergenerational gains. (3) The 5% discount rate and 35-year horizon are standard, neither aggressive nor generous. The one assumption that pushes the other way (that a girl exiting NEET earns the median wage immediately) makes the estimate an upper bound on what is realistically and immediately recoverable; but the deliberately low return benchmark more than offsets this, so the overall figure is best read as conservative.

A.4 Sources for budget comparisons

Figure	Value (NPR)	Fiscal year	Source
Cost of inaction (one cohort)	8.1 billion	—	This analysis, NLSS IV 2022/23 (NSO Nepal)
Health-sector budget (total MoHP)	86.24 billion	2024/25	MoF budget, reported by Kathmandu Post, 29 May 2024
Social security allowance budget	106 billion	2022/23	UNICEF Nepal, Social Protection Budget Brief Update FY2022/23
Education-sector budget (MoEST)	203.66 billion	2024/25	MoF Budget Speech FY2024/25, para 176
Provincial GDP (all 7 provinces)	various	2023/24	National Statistics Office (NSO) Nepal, provincial accounts 2023/24

Budget source links:

• Health budget (Kathmandu Post, 29 May 2024): https://kathmandupost.com/health/2024/05/29/government-sets-aside-a-meagre-4-6-percent-of-the-annual-budget-for-the-health-sector

• Health budget cross-check (Nepal News, 27 Oct 2025): https://english.nepalnews.com/s/health/as-ncds-surge-nepals-health-budget-falls/

• Social security allowances (UNICEF Nepal SP Budget Brief FY2022/23): https://www.unicef.org/nepal/media/18096/file/Social%20Protection%20Budget%20Brief%20Update:%20FY%202022-23.pdf

• Education budget (MoF Budget Speech FY2024/25): https://demo.mof.gov.np/uploads/document/file/1724825318_Budget%20Speech%202024_25%20English%20FV.pdf

• Total national budget cross-check (Nepal Economic Forum): https://nepaleconomicforum.org/whats-inside-fiscal-budget-2081-82-key-highlights-and-programs/

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